

Bilinear Forms

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Problem 5a. Now we turn to a real symmetric matrix A . What we will show is that we can find orthogonal P so that $P^t A P = \text{diagonal}$. You would think the Th above is a step in this direction. I do not see how to use it. If you do, of course make some noise. Instead, let $[\]$ be standard dot product. Note A symmetric is equiv to $[Av, w] = [v, Aw]$ for all v, w .

Now to prove the theorem, we want a coordinate free formulation of symmetric matrix.

Thm 2: Let V be a real vector space, and $[\]$ a symmetric form which is "positive def", ie $[v, v] > 0$ for $v \neq 0$. Let $A: V \rightarrow V$ be linear with $[Av, w] = [v, Aw]$ for all v, w (this is called "self adjoint").

Then there is a basis, orthonormal w/r to $[\]$, consisting of eigenvectors. Lemma: all eigenvalues of A are real.

Assume the lemma, prove the Thm 2 by induction on the dimension of V . Deduce from it the original statement (that a real symmetric matrix is orthogonally diagonalizable).

Proof. (Base Case) ($k = 1$): Suppose V is a real vector space with dimension 1. Thus for some $v \in V \ni v \neq 0$, $\dim \text{span}(v) = \dim V \implies \text{span}(v) = V$. Thus $\forall w \in V$, $w = av$. For all $A \in \text{Hom}_{\mathbb{R}}(\mathbb{R}, \mathbb{R})$, $Av \in V = \text{span}(v)$, thus $Av = \lambda v$ thus by definition v is an eigenvector. $[\frac{1}{\sqrt{[v, v]}}v, \frac{1}{\sqrt{[v, v]}}v] = \frac{1}{[v, v]}[v, v] = 1$ thus $\frac{1}{\sqrt{[v, v]}}v$ is normal. $\forall w \in V$, since V is a one dimensional real vector space and $v \neq 0$ thus $\dim \text{span}(v) = 1$. Thus $\text{span}(v) \subset V$ and $\dim V = \dim \text{span}(v) \implies v$ is a basis of V . Thus there exists a orthonormal eigenbasis for V .

Induction Hypothesis: Suppose the theorem is true for all real vector spaces with dimension $k - 1$.

Inductive Step: Let V be a real vector space with dimension k , and $[\]$ be a symmetric form which is positive definite. Let $A: V \rightarrow V$ be a linear map which is self adjoint with $[\]$. Because in complex vector space V , any linear map $A: V \rightarrow V$ is diagonalizable since χ_A has all its roots in A . Thus A has all its eigenvalues in $\mathbb{C}$. But all the eigenvalues of A are real by the Lemma. Thus there exists a real eigenvalue λ . Thus the linear map $B = A - \lambda I$ is real, since A and I are real linear maps. To calculate the $\ker(A - \lambda I)$ one must do gaussian elimination after choosing a basis for V , both of which do not add any complex dependence. Thus $\ker(A - \lambda I)$ is nonzero and has real vectors. Thus $\exists$ a eigenvector $v \in \ker(A - \lambda I) \ni v \neq 0$ and v is real. Furthermore since $[\] > 0$ (positive definite), $[v, v] \neq 0$ since $v \neq 0$. WLOG v is normal or $[v, v] = 1$.

$\forall w \in \text{span}(v) \cap v^\perp$, $[v, w] = 0$, since $w \in v^\perp$. But since $w \in \text{span}(v)$, $w = av$ for some $a \in \mathbb{R}$. Thus $a[v, v] = [v, av] = [v, w] = 0$. Since $[v, v] \neq 0$ because $[\]$ is positive definite, $a = 0$ and $w = 0$. Thus $\text{span}(v) \cap v^\perp = 0$. $\text{span}(v) \oplus v^\perp \rightarrow V$ is injective.

Let $f: W \rightarrow \mathbb{R}$ which takes $w \rightarrow [v, w]$. $v^\perp := \ker(f) = \{w \in V : [v, w] = 0\}$. $f(v) = 1$ and $\forall x \in \mathbb{R}$, $f(xv) = x$. Thus $f_*(V) = \mathbb{R}$ and $\dim f_*(V) = 1$. By rank nullity, $\dim V = \dim f_*(V) + \dim v^\perp \implies \dim v^\perp = \dim V - \dim \mathbb{R} = \dim V - 1 = k - 1$. Since $\text{span}(v) \oplus v^\perp \rightarrow V$ is injective and $\dim \text{span}(v) + \dim v^\perp = 1 + \dim V - 1 = \dim V$, the map is surjective and thus is a bijection. $\forall (v, w), (x, y) \in V \oplus W$ and $a, b \in \mathbb{R}$, $a(v, w) + b(x, y) \cong av + bw + ax + by \cong (av + bx, aw + by)$ thus the map is also linear and thus is an isomorphism.

$\forall u, w \in v^\perp$, $[u, w] = [w, u]$ because $[]$ is symmetric. Furthermore, $[Au, w] = [Aw, u]$, by the fact that A is self adjoint to the symmetric form. Furthermore, $\forall w \in v^\perp$ $[v, Aw] = [Av, w]$ since A is self adjoint to the symmetric form. Thus $[v, Av] = [Av, w] = [\lambda v, w] = \lambda[v, w] = 0$. Thus $Aw \in v^\perp$. Thus $Av^{perp} \subset v^\perp$. Thus $A|_{v^\perp} : v^\perp \rightarrow v^\perp$. Restrict A and the Bilinear form to $v^\perp$. By the inductive Hypothesis exists a orthonormal eigenbasis, $b_1, \dots, b_{k-1}$. By definition of $v^\perp$, $[v, b_i] = 0$ and by definition of orthonormal eigenbasis, $[b_i, b_j] = \delta(i, j)$. Thus $v, b_1, \dots, b_{k-1}$ is a orthonormal eigenbasis for V . $\square$

Problem 5b. *It remains to prove the lemma. Choose an orthonormal basis (with respect to $[,]$), so that $A : V \rightarrow V$ becomes a matrix. Show this matrix is symmetric. So the lemma is equivalent to showing a real symmetric matrix has real eigenvalues. More generally, show that if B is a square complex matrix with $B^t = \bar{B} :=$ complex conjugate, then for $[,]$ the standard hermitian dot product, $[Bv, w] = [v, Bw]$, and show all the eigenvalues are real.*

Claim 5bi. *A is symmetric.*

Proof. Since a orthonormal basis $b_1, \dots, b_k$ is chosen. $[b_i, b_j] = \delta(i, j)$. Additionally under this basis, we can write, $[x, y] = x^T M y$ for some matrix M . $[b_i, b_i] = b_i^T M b_i = 1 \implies M_{i,i} = 1$, and if $i \neq j$, $[b_i, b_j] = b_i^T M b_j = 0 \implies M_{i,j} = 0$. So $M = I$.

$$[Ab_i, b_j] = b_i^T A^T I b_j = b_i^T I A b_j = [b_i, A b_j] \implies b_i A^T b_j = b_i^T A b_j \implies A_{i,j} = A_{j,i} \implies A = A^T.$$

Thus A is symmetric. $\square$

Claim 5bii. *Show that if B is a square complex matrix with $B^t = \bar{B} :=$ complex conjugate, then for $[,]$ the standard hermitian dot product, $[Bv, w] = [v, Bw]$, and show all the eigenvalues are real.*

Proof. Let λ be a eigenvalue of B and v be the associated eigenvector. By the properties of the standard hermetian dot product where $\forall w \in V$, $[w, w] = 0 \iff w = 0$ otherwise $[w, w] = w^T \bar{w} > 0$, thus $[v, v] > 0$ since $v \neq 0$. $[Bv, v] = [v, Bv] \implies [\lambda v, v] = [v, \lambda v] \implies (\lambda v)^T \bar{v} = v^T \bar{\lambda} v \implies \lambda v^T v = \bar{\lambda} v^T v$. Since $v^T v = [v, v] > 0$, it means $\lambda = \bar{\lambda} \implies \lambda \in \mathbb{R}$. $\forall v, w \in V$, $[Bv, w] = (Bv)^T \bar{w} = v^T B^T \bar{w} = v^T \bar{B} \bar{w} = [v, Bw]$. $\square$

Claim 5biii. *Lemma is true*

Proof. $A = \bar{A} = A^T \implies$ all eigenvalues of A are real by 5bii. $\square$

Problem 6. *Note the main thing in proving Thm 2, is that A behaves well with respect to a positive def form. For a generalization of Thm 2, see the "Spectral Theorem" in Artin chapter 8.*

Solution. Not a question $\blacksquare$

Problem 7. *Let $\langle, \rangle$ be a symmetric form on a finite dim real V . Show there is a basis $b_1, \dots, b_n$ with $\langle b_i, b_j \rangle = 0$ for $i \neq j$, and $\langle b_i, b_i \rangle$ either 1, -1 or 0. Show that the numbers p, q, r of 1s, -1s and zeros are invariants of $\langle, \rangle$ (ie the same for any two such bases). Show first that k is the dim of the kernel of $\langle, \rangle$ (define what that means, there is only one reasonable definition). Let $V \rightarrow V/K$ be quotient by the kernel, show that $\langle, \rangle$ induces a form on V/K . Let D be a subspace on which $\langle, \rangle$ is definite (either positive def, or negative definite). Show $D \cap K = 0$. Conclude its enough to prove the theorem when $K=0$. Let P and Q be subspaces on which $\langle, \rangle$ is positive and negative. Show $P \cap Q = 0$. Now prove the invariance.*

Claim 7a. Show there is a basis $b_1, \dots, b_n$ with $\langle b_i, b_j \rangle = 0$ for $i \neq j$, and $\langle b_i, b_i \rangle$ either 1, -1 or 0.

Proof. Define $|x| = \sqrt{x^2}$

By 5, $\exists$ basis $v_1, \dots, v_n \ni \langle v_i, v_j \rangle = 0$ for $i \neq j$. Create a basis $b_1, \dots, b_j$ where $b_i = \frac{1}{\sqrt{|\langle v_i, v_i \rangle|}} v_i$ if $\langle v_i, v_i \rangle \neq 0$ else $b_i = v_i$. If $\langle v_i, v_i \rangle \neq 0$, $\langle b_i, b_i \rangle = \langle \frac{1}{\sqrt{|\langle v_i, v_i \rangle|}} v_i, \frac{1}{\sqrt{|\langle v_i, v_i \rangle|}} v_i \rangle = \frac{1}{|\langle v_i, v_i \rangle|} \langle v_i, v_i \rangle = \pm 1$. If $\langle v_i, v_i \rangle = 0 \implies \langle b_i, b_i \rangle = 0$. So we have a created a basis where $\langle b_i, b_j \rangle = 0$ if $i \neq j$ and $\langle b_i, b_i \rangle = 0, 1, -1$. $\square$

Claim 7b. Show first that k is the dim of the kernel of $\langle, \rangle$ (define what that means, there is only one reasonable definition).

Proof. Let $u_1, \dots, u_p, v_1, \dots, v_q, w_1, \dots, w_r$ be basis of V where the Bilinear form evaluated on any two different basis vectors is 0 and $\langle u_i, u_i \rangle = 1$ and $\langle v_i, v_i \rangle = -1$ and $\langle w_i, w_i \rangle = 0$. $W = \{w_i : i = [1, \dots, r]\}$ is the set of basis vectors where $\langle w_i, w_i \rangle = 0$. $r = |W|$.

$K = \ker \langle, \rangle := \{x | \langle x, v \rangle = 0, \forall v \in V\}$. $\forall w_i, \langle w_i, \sum_{i=1}^p a_i u_i + \sum_{i=1}^q b_i v_i + \sum_{i=1}^r c_i w_i \rangle = \sum_{i=1}^r c_i \langle w_i, w_i \rangle = 0 \implies w_i \in K$. Thus $W \subset K \implies \text{span}(W) \subseteq K$.

$\forall v \in K, v = \sum_{i=1}^p a_i u_i + \sum_{i=1}^q b_i v_i + \sum_{i=1}^r c_i w_i$. $\langle v, u_i \rangle = a_i \langle u_i, u_i \rangle = 0 \implies a_i = 0$ ($\langle u_i, u_i \rangle = 1$) and $\langle v, v_i \rangle = b_i \langle v_i, v_i \rangle = 0 \implies b_i = 0$ ($\langle v_i, v_i \rangle = -1$). Thus $v \in \text{span}(W)$. Thus $\text{span}(W) \cong K$. $\dim K = \dim \text{span}(W) = r$

So any basis vector $b_i \ni Q(b_i) = 0 \implies b_i \in \ker Q$. Thus $k = \dim \ker Q$. $\square$

Claim 7c. $\langle, \rangle$ induces a form on V/K

Proof. $\forall vK, wK \in V/K, [vK, wK] = \langle v, w \rangle$.

$\forall k_1, k_2 \in K$ and $\forall v, w \in V, v + k_1 \in vK$ and $w + k_2 \in wK, [(v + k_1)K, (w + k_2)K] = \langle v + k_1, w + k_2 \rangle = \langle v, w \rangle + \langle v, k_2 \rangle + \langle k_1, w \rangle + \langle k_1, k_2 \rangle = \langle v, w \rangle = [vK, wK]$. Form is well defined.

$\langle, \rangle$ factors through $[,]$ form and thus induces it on V/K . $\square$

Claim 7d. Let D be a subspace on which j, j is definite (either positive def, or negative definite). Show $D \cap K = 0$.

Proof. $\forall v \in D \cap K$, by definition, $v \in D$ and $v \in K$. So, $Q(v) = 0$. By definition of definite, if $v \neq 0 \implies Q(v) \neq 0$, else $Q(v) = 0$. Since $Q(v) = 0 \implies v = 0$. So $D \cap K = 0$ $\square$

Claim 7e. Let P and N be subspaces on which $\langle, \rangle$ is positive and negative. Show $P \cap N = 0$.

Proof. $\forall v \in P \cap N$, by definition, $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle \leq 0$. Thus $\langle v, v \rangle = 0$. But $\forall a \in P$ and $\forall b \in N, \langle a, a \rangle = 0 \iff a = 0$ and $\langle b, b \rangle = 0 \iff b = 0$. So $v = 0$. So $P \cap N = 0$ $\square$

We can consider $K = 0$ because otherwise we can consider V/K and its induced form for the theorem WLOG.

Claim 7f. Prove the invariance.

Proof. Natural addition map $P \oplus N \rightarrow V$ is injective because $P \cap N = 0$ as proven above. Let B_1 and B_2 be two orthonormal basis. Let P_1 be the subspace formed by the positive definite basis vectors of B_1 and N_1 be the subspace of the negative definite basis vectors of the same. Similarly let P_2 be the subspace formed by the positive definite basis vectors of B_2 and N_2 be the subspace of the negative definite vectors of B_2 .

So $P_1 \cap N_1 = 0$, $P_1 \cap N_2 = 0$, $P_2 \cap N_2 = 0$, $P_2 \cap N_1 = 0$ as proven above. Because if $v \in P_i \cap N_j \implies 0 \geq \langle v, v \rangle \leq 0 \implies \langle v, v \rangle = 0$ and $K = 0 \implies v = 0$.

$a : P_1 \subset V$ and let $b : V \rightarrow V/N_2$ be the natural quotient map. $\ker[b \circ a] = P_1 \cap N_2 = 0 \implies q \circ p$ is injective and $P_1 \subseteq P_2 = V/N_2$. Argument is symmetric for $P_2 \subseteq P_1$. $c : P_2 \subset V$ and let $d : V \rightarrow V/N_1$. $\ker[d \circ c] = P_2 \cap N_1 = 0 \implies d \circ c$ is injective and $P_2 \subseteq P_1 = V/N_1$. Thus $P_1 = P_2 \implies N_1 = N_2$. Thus the number of +1 and -1 is invariant. $\square$

Problem 8a. We say a bilinear form on a real vector space is non-degenerate if there are no zeros. Show this is the same as having an invertible matrix, in any basis

Proof. Suppose $\langle, \rangle$ is nondegenerate. By 7a, $\exists$ a orthonormal basis where $i \neq j \implies \langle b_i, b_j \rangle = 0$ and $\langle b_i, b_i \rangle = 0, 1, -1$. Since $\langle, \rangle$ is nondegenerate there is no basis vector $b \ni \langle b, b \rangle = 0$. Let M be the matrix associated with the Bilinear form under this basis. So $\langle x, y \rangle = x^T M y$. M would have 0s on the off diagonal entries since $\langle b_i, b_j \rangle = 0$ for $i \neq j$ and ± 1 on the diagonal entries. Since the determinant of a diagonal matrix is the product of the diagonal entries and no diagonal entries are 0, $\det(M) \neq 0$. Let P be a matrix to change the basis. By def, $\det(P) \neq 0$. So $\det(PMP^{-1}) = \det(P) \det(M) \frac{1}{\det(P)} = \det(M) \neq 0$ and $\det(P^T M P) = \det(P)^2 \det(M) \neq 0$. Thus under any basis the matrix will be invertible. $\square$

Problem 8b. Show that for any real matrix A (not necessarily square) $A^T A$ is symmetric, and positive semi-definite, and moreover positive definite iff A is invertible (in particular square). Now suppose B is semi-definite and symmetric. Show $B = A^T A$ for some square A .

Proof. Let A be a real matrix. $(A^T A)^T = A^T A^{TT} = A^T A \implies A^T A = (A^T A)^T$. So A is symmetric. Let $[\cdot]_{A^T A}$ be the Bilinear form associated with $A^T A$. By 7, there is a basis where $i \neq j \implies [b_i, b_j]_{A^T A} = 0$ and $[b_i, b_i]_{A^T A} = 1, -1, 0$. Thus $[b_i, b_i]_{A^T A} = b_i^T A^T A b_i = (A b_i)^T \cdot (A b_i) \geq 0$ by standard dot product. So $[b_i, b_i]_{A^T A} \geq 0$, thus $A^T A$ is positive semidefinite.

Suppose A is positive definite, then $[b_i, b_i]_{A^T A} = 1 = b_i^T A^T A b_i \implies [A^T A]_{i,i} = 1$. Thus offdiagonals of $A^T A$ is 0 and diagonal entries are 1. Thus $A^T A = I$. $\forall v \in \ker(A^T A)$, $A^T A v = 0 \implies \forall w \in V$, $[w, v]_{A^T A} = 0 = w^T A^T A v = (A w) \cdot (A v) = 0$. For $w = v$, $(A v) \cdot (A v) = 0 \implies A v = 0$ (by standard dot product). Thus $v \in \ker A$. Thus $\ker A^T A \subset \ker A$. $\forall v \in \ker A \implies A v = 0 \implies A^T A v = 0 \implies v \in \ker A^T A \implies \ker A \subseteq \ker A^T A \implies \ker A^T A = \ker A = \ker I = 0$. Thus A is invertible.

Suppose A is invertible. $\det(A^T A) = \det(A)^2 \neq 0$ thus no diagonal entries can be 0. So $[b_i, b_i] = b_i^T A^T A b_i = [A^T A]_{i,i} \neq 0$. So $A^T A$ is positive definite.

Now suppose B is semi-definite and symmetric. Under the orthonormal basis of the bilinear form $\langle v, w \rangle = v^T B w$ (Bilinear form between basis is 0 and Bilinear form of basis with itself is 0 or 1). The basis vectors can be ordered with those which are positive and which are 0. So B is diagonal under

this basis. So $B = \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 0 & \\ & & & \ddots \end{bmatrix}$, let $A = B$. $A^T A = B$. $\square$

Problem 9. Let $\langle, \rangle$ be a symmetric non-degenerate form on a finite dim complex vector space. Show there is a basis in which the associated matrix is the identity. If we want to measure distance, then the right notion is "Hermitian symmetric". This means $\langle, \rangle$ is linear in the first variable, conjugate linear

in the second, and $\langle x, y \rangle = \langle y, x \rangle$. Show this holds iff the associated matrix satisfies $A^T = \bar{A}$. Now copy the above to prove that given a hermitian symmetric matrix, there is a unitary matrix g such that $gAg^{-1} = \text{diagonal}$. Unitary means that g preserves the standard (hermitian symmetric) dot product on $\mathbb{C}^n$. equivalently $g^t \bar{g} = I$.

Claim 9a. Let $\langle, \rangle$ be a symmetric non-degenerate form on a finite dim complex vector space. Show there is a basis in which the associated matrix is the identity.

Proof. By 7a, we can create a basis $b_1, \dots, b_k$ where $\langle b_i, b_j \rangle = 0$ if $i \neq j$ and $\langle b_i, b_i \rangle = 0, 1, -1$. Since $\langle, \rangle$ is nondegenerate, $\forall b_i, \langle b_i, b_i \rangle \neq 0$. Create a basis $v_1, \dots, v_k$, where $v_i = b_i \iff \langle b_i, b_i \rangle = 1$ else $v_i = ib_i \iff \langle b_i, b_i \rangle = -1$. Thus if $\langle b_i, b_i \rangle = 1 \implies \langle v_i, v_i \rangle = 1$. If $\langle b_i, b_i \rangle = -1 \implies \langle v_i, v_i \rangle = i^2 \langle b_i, b_i \rangle = 1$. Thus the associated matrix has 1s on the diagonal. For all $i \neq j$, $\langle v_i, v_j \rangle = \langle b_i, b_j \rangle = 0$, thus the associated matrix has 0 off the diagonal. Thus the associated matrix is I by definition. Since $\langle b_i, b_j \rangle = b_i M b_j = \delta(i, j) \implies M = [\delta(i, j)]_{i,j} = I$ $\square$

Claim 9b. Show this holds iff the associated matrix satisfies $A^T = \bar{A}$.

Proof. $\forall x, y \in V, \langle x, y \rangle = x^T A \bar{y} = \langle y, x \rangle = y^T \bar{A} x = \bar{y}^T A x \implies (x^T A \bar{y})^T = \bar{y}^T A^T x = \bar{y}^T \bar{A} x \implies A^T = \bar{A}$ $\square$

Claim 9c. Let V be the complex vector space and $[,]$ be a positive definite hermitian symmetric bilinear form. Let $A : V \rightarrow V \ni \forall v, w \in V, [Av, w] = [v, Aw]$. There exists a orthonormal eigenvector basis wrt to $[,]$ and A .

Proof. Base Case ($\dim V = 1$): Let v be a eigenvector. WLOG, v is normal. v is a orthonormal eigenbasis.

Let the claim be true for all complex vectorspaces with $\dim V = k - 1$.

Suppose V is a complex vector space with $\dim V = k$ and let λ be its corresponding eigenvalue. Let v be a eigenvector of A . WLOG v is normal. $f : V \rightarrow \mathbb{C} \ni w \rightarrow [v, w]$. $v^\perp := \ker(f)$. $\dim \ker f = \dim V - \dim f_*(V) = \dim V - 1$. $\forall w \in \text{span}(v) \cap v^\perp \implies [w, w] = 0 \implies w = 0 \implies \text{span}(v) \cap v^\perp = 0$. $\dim \text{span}(v) + \dim \ker f = \dim V$. Thus $\text{span}(v) \oplus v^\perp \cong V$. $\forall w \in \text{span}(v), w = av, a \in \mathbb{C}, Aw = Aav = a\lambda v \in \text{span}(v)$. So $A_*(\text{span}(v)) \subseteq \text{span}(v)$. $\forall w \in v^\perp, [Aw, v] = [w, Av] = \lambda[w, v] = 0 \implies A_*(v^\perp) \subset v^\perp$. Restrict A and the $[,]$ to $v^\perp$, by inductive hypothesis $\exists$ orthonormal eigenbasis of $v^\perp$, $b_1, \dots, b_{k-1}$. By definition $[b_i, b_j] = 0$ for $i \neq j$ and $[v, b_i] = 0$. Thus $v, b_1, \dots, b_{k-1}$ is orthonormal eigenbasis for V . $\square$

Claim 9d. Now prove that given a hermitian symmetric matrix, there is a unitary matrix g such that $gAg^{-1} = \text{diagonal}$.

Proof. Let $\langle, \rangle$ be the standard hermitian dot product. Since $A^T = \bar{A}$, $\forall v, w \in \mathbb{C}^n, [Av, w] = v^T A^T \bar{w} = v^T \bar{A} \bar{w} = [v, Aw]$. By the 9c, there is a orthonormal eigenbasis wrt to A and standard hermitian dot product, $b_1, \dots, b_n$. Let $P \in GL_n(\mathbb{C})$ be the corresponding automorphism which sends old basis to new orthonormal eigenbasis. $[b_i, b_j] = [Pv_i, Pv_j] = v_i^T P^T \bar{P} v_j = \delta(i, j) \implies P^T \bar{P} = I$. $PAP^{-1} = \text{diagonal}$ because of eigenbasis basis. $\square$

Problem 10. What does it mean that the Hessian is positive definite, or negative definite? Or semi-definite?

Solution. If the Hessian at a critical point is positive definite, the critical point is a local minimum of the function (function is concave up). If it is negative definite at the critical point, the critical point is a local maxima of the function (function is concave down). If it is semi-definite, then the critical point is inconclusive, over some subspace the critical point can be a local minimum (concave up), over the complementary subspace it could be flat. ■

Problem 11. Let $j, \dot{\bar{}}$ be a skew form on a finite dim vector space over a field of characteristic other than 2. Assume $j, \dot{\bar{}}$ is non-degenerate (meaning, no kernel). Then there is a basis $e_1, f_1, \dots, e_k, f_k$ with $\langle e_i, f_j \rangle = \langle e_i, e_j \rangle = \langle f_i, f_j \rangle = 0$ for all $i \neq j$, and $\langle e_i, f_i \rangle = 1$.

Proof. (Base Case): Let V be a vector space over a field of characteristic other than 2, where $\dim V = 2$. Let $\langle, \rangle$ be a nondegenerate skew form on the V . $e \in V \ni e \neq 0$. By nondegeneracy, $\exists f' \in V \ni \langle e, f' \rangle \neq 0$. $f := \frac{1}{\langle e, f' \rangle} f'$, so $\langle e, f \rangle = \frac{1}{\langle e, f' \rangle} \langle e, f' \rangle = 1$. e, f are the basis which satisfies the theorem.

(Inductive Hypothesis): Let the theorem be true for any vector space V over a field F ($\text{char} F \neq 2$), where $\dim V < k$.

Inductive Step: Let V be a vector space over a field of characteristic other than 2, where $\dim V = k$. Let $\langle, \rangle$ be a nondegenerate skew form on the V . $e \in V \ni e \neq 0$. By nondegeneracy, $\exists f' \in V \ni \langle e, f' \rangle \neq 0$. $f := \frac{1}{\langle e, f' \rangle} f'$, so $\langle e, f \rangle = \frac{1}{\langle e, f' \rangle} \langle e, f' \rangle = 1$

Let $S = \text{span}(e, f)$. Let $\alpha : V \rightarrow F^2$, where $v \mapsto \begin{bmatrix} \langle e, v \rangle \\ \langle f, v \rangle \end{bmatrix}$. Let $S^\perp = \{a \in V \ni \langle a, b \rangle = 0, \forall b \in S\} = \ker \alpha$.

$\forall v \in S \cap S^\perp$, $v = ae + bf$, $\langle v, e \rangle = a\langle e, e \rangle - b\langle e, f \rangle = -b\langle e, f \rangle = 0 \implies b = 0$. $\langle v, f \rangle = a\langle e, f \rangle + b\langle f, f \rangle = a\langle e, f \rangle = 0 \implies a = 0$. Thus $v = 0$. Thus $S \cap S^\perp = 0$.

By Rank Nullity, $\dim \ker \alpha = \dim V - \dim \alpha_*(V)$. $\alpha(e) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $\alpha(f) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, thus $\alpha(e)$ and $\alpha(f)$ are linearly independent. $2 \leq \dim \alpha_*(V)$. Thus $\dim S^\perp = \dim \ker \alpha \geq \dim V - 2$. Thus $\dim V \leq \dim S^\perp + \dim S \implies \dim S^\perp + \dim S = \dim V \implies S^\perp \oplus S \rightarrow V$ is an isomorphism. $\forall v \in S^\perp$, $\langle v, s \rangle = 0, \forall s \in S$. However by nondegeneracy, $\exists w \in V \ni \langle v, w \rangle \neq 0$, by previous statement $w \notin S \implies w \in V/S = S^\perp$. Thus the restriction of $\langle, \rangle$ to $S^\perp$ must be nondegenerate. By induction, $\exists$ basis $e_1, f_1, \dots, e_k, f_k$ with $\langle e_i, f_j \rangle = \langle e_i, e_j \rangle = \langle f_i, f_j \rangle = 0$ for all $i \neq j$, and $\langle e_i, f_i \rangle = 1$. $\langle e, e_i \rangle = 0 = \langle e, f_i \rangle = \langle f, f_i \rangle$ (by definition of $S^\perp$). Thus the basis $e, f, e_1, f_1, \dots, e_k, f_k$ is a basis which satisfies the theorem. □

Problem 12. Any invertible skew symmetric matrix, over a field of char not equal to two, must have even size (ie if its $n \times n$, then n is even)

Proof. $A^T = -A$. $\det(A) = \det(A^T) = \det(-A) = (-1)^n \det(A) \implies (-1)^n = 1$. Assume the contrary, n is odd and $(-1)^n = -1$. So $n = 2k + 1$ for some $k \in \mathbb{Z}$. $(-1)^{2k+1} = (-1)^{2k}(-1) = ((-1)^2)^k(-1) = 1^k(-1) = 1(-1) = -1$. Thus a contradiction exists and n must be even. □

Problem 13. Any two invertible skew symmetric $n \times n$ matrices over a field of char $\neq 2$ are "equivalent" for the action of $GL_n(F)$ on $n \times n$ matrices by $g \cdot M := gMg^t$, ie show that the invertible skew symmetric matrices are a single orbit. What are the other orbits for the action on skew symmetric matrices? How about $GL_n(\mathbb{R})$ acting on symmetric real $n \times n$ matrices?

Proof. Let K be the set of invertible skew symmetric $n \times n$ matrices over $\text{char} \neq 2$.

$\forall A, B \in K$. Let $\langle, \rangle$ be a skew symmetric bilinear form where $\forall v, w \in F^n$, $\langle v, w \rangle = v^T A w$. Let $[,]$ be a skew symmetric Bilinear form where $\forall v, w \in F^n$, $[v, w] = v^T B w$. Let $B = s_1, \dots, s_n$ be the standard basis. $\langle s_i, s_j \rangle = s_i^T A s_j = A_{i,j} = -A_{j,i} = s_j^T A s_i = \langle s_j, s_i \rangle$. $[s_i, s_j] = B_{i,j} = -B_{j,i} = [s_j, s_i]$. $\langle, \rangle$ and $[,]$ are nondegenerate since A and B are invertible.

By 11, $\exists$ basis $B_1 = e_1, f_1, \dots, e_k, f_k$ with $\langle e_i, f_j \rangle = \langle e_i, e_j \rangle = \langle f_i, f_j \rangle$ for all $i \neq j$, and $\langle e_i, f_i \rangle = 1$ and $\exists$ basis $B_2 = a_1, b_1, \dots, a_k, b_k$ with $[a_i, b_j] = [a_i, a_j] = [b_i, b_j] = 0$ for all $i \neq j$, and $[a_i, b_i] = 1$. Let $C \in GL_n(F)$ be the linear transformation which transforms basis B to B_1 . Let $D \in GL_n(F)$ be the linear transformation which transforms basis B to B_2 .

$$\begin{aligned} \langle e_i, f_j \rangle &= \langle C s_{2i-1}, C s_{2j} \rangle = s_{2i-1}^T C^T A C s_{2j} = [C^T A C]_{2i-1, 2j} = 0, \\ \langle e_i, e_j \rangle &= \langle C s_{2i-1}, C s_{2j-1} \rangle = s_{2i-1}^T C^T A C s_{2j-1} = [C^T A C]_{2i-1, 2j-1} = 0 \\ \langle f_i, f_j \rangle &= [C^T A C]_{2i, 2j} = 0 \\ \langle e_i, e_i \rangle &= 0 = [C^T A C]_{2i-1, 2i-1} = 0. \\ \langle f_i, f_j \rangle &= 0 = [C^T A C]_{2i, 2i} = 0 \\ \langle e_i, f_i \rangle &= 1 = [C^T A C]_{2i-1, 2i} = 1 \\ \langle f_i, e_i \rangle &= -1 = [C^T A C]_{2i, 2i-1} = 1 \end{aligned}$$

Let S_n , be the $n \times n$ matrix which has 1s on the diagonal above the main diagonal, and -1s on the diagonal under the main diagonal and 0s everywhere else.

$$\text{So } C^T A C = \begin{bmatrix} S_2 & & \\ & S_2 & \\ & & \ddots \end{bmatrix} = G$$

$$\begin{aligned} \text{Similarly, } [a_i, b_j] &= [D s_{2i-1}, D s_{2j}] = s_{2i-1}^T D^T B D s_{2j} = [D^T B D]_{2i-1, 2j} = 0, \\ [a_i, a_j] &= D s_{2i-1}^T D s_{2j-1} = s_{2i-1}^T D^T B D s_{2j-1} = [D^T B D]_{2i-1, 2j-1} = 0 \\ b_i, b_j &= [D^T B D]_{2i, 2j} = 0 \\ a_i, a_i &= 0 = [D^T B D]_{2i-1, 2i-1} = 0. \\ b_i, b_i &= 0 = [D^T B D]_{2i, 2i} = 0 \\ a_i, b_i &= 1 = [D^T B D]_{2i-1, 2i} = 1 \\ b_i, a_i &= -1 = [D^T B D]_{2i, 2i-1} = 1 \end{aligned}$$

$$\text{Thus } D^T B D = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots \\ -1 & 0 & 1 & \ddots & \ddots \\ 0 & -1 & 0 & 1 & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix} = G.$$

Thus $D^T B D = C^T A C$. Let $H = D^{-1}$. $H^T D^T B D H = B = H^T C^T A C H = (CH)^T A C H$. Thus all elements of K are in the same orbit.

Let $\text{Skew}(F^n)$ be the set of skew symmetric matrices. $\forall A \in F^n$, $F^n \setminus \ker A \cup \{0\} \oplus \ker A \cong V$. Apply theorem to $F^n \setminus \ker A \cup \{0\}$ for the skew symmetric bilinear form, $\angle v, w_A = v^T A w$. Thus there exists a basis of $F^n \setminus \ker A \cup \{0\}$, $e_1, f_1, \dots, e_k, f_k$ with $\langle e_i, f_j \rangle = \langle e_i, e_j \rangle = \langle f_i, f_j \rangle$ for all $i \neq j$, and $\langle e_i, f_i \rangle = 1$. Add the basis of $\ker A$, $k_1, \dots, k_n$. $\ker \langle, \rangle_A \subset \ker A$. So under the change of basis automorphism P . $P^T A P = \begin{bmatrix} S_{\dim V - \dim \ker A} & 0 \\ 0 & 0 \end{bmatrix}$.

$$\begin{aligned} \forall A, B \in GL_n(F)(A), \text{ by the above } \exists P, Q \in GL_n(F)(A), P^T A P &= \begin{bmatrix} S_{n - \dim \ker A} & 0 \\ 0 & 0 \end{bmatrix} = A' \text{ and} \\ Q^T B Q &= \begin{bmatrix} S_{n - \dim \ker B} & 0 \\ 0 & 0 \end{bmatrix} = B'. \text{ Since } A', B' \in GL_n(F)(A), \exists P' \in GL_n(F), P' T \begin{bmatrix} S_{n - \dim \ker A} & 0 \\ 0 & 0 \end{bmatrix} P = \end{aligned}$$

$$\begin{bmatrix} S_{n-\dim \ker B} & 0 \\ 0 & 0 \end{bmatrix}. \text{ By 13B, } \dim \ker \begin{bmatrix} S_{n-\dim \ker A} & 0 \\ 0 & 0 \end{bmatrix} = \dim \ker A = \dim \ker B = \dim \ker \begin{bmatrix} S_{n-\dim \ker B} & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus skew symmetric matrices in the same orbit have the same kernel.

Let A and B be skew symmetric matrices where $\ker A = \ker B = k$. $\exists P, Q \in GL_n(F) \ni P^T A P = \begin{bmatrix} S_{n-k} & 0 \\ 0 & 0 \end{bmatrix} = Q^T B Q \implies A, B \in GL_n(F)(A)$. $\square$

Problem 13b. How about $GL_n(\mathbb{R})$ acting on symmetric real $n \times n$ matrices?

Proof. $GL_n(\mathbb{R})$ is acting on the set of real symmetric matrices. Let $[\cdot, \cdot]$ be the standard dot product.

$\forall A, B \in GL_n(\mathbb{R})(A)$, by definition A, B are self adjoint to the dot product. By 7, $\exists$ a eigenbasis for A and B where the dot product of 2 different basis elements is 0 and the dot product of a basis element with itself is 0, -1, or 1. Let $u_1, \dots, u_{n_a}, v_1, \dots, v_{m_a}, w_1, \dots, w_{k_a}$ be the eigenbasis for A where $\langle u_i, u_i \rangle = 1$, $\langle v_i, v_i \rangle = -1$ and $\langle w_i, w_i \rangle = 0$. Similarly let $x_1, \dots, x_{n_b}, y_1, \dots, y_{m_b}, z_1, \dots, z_{k_b}$ be the

corresponding eigenbasis for B . Thus there $\exists P, Q \in GL_n(\mathbb{R}) \ni P^T A P = M = \begin{bmatrix} I_{n_a} & 0 & 0 \\ 0 & -I_{m_a} & 0 \\ 0 & 0 & 0_{k_a} \end{bmatrix}$,

$Q^T B Q = N = \begin{bmatrix} I_{n_b} & 0 & 0 \\ 0 & -I_{m_b} & 0 \\ 0 & 0 & 0_{k_b} \end{bmatrix}$. $N, M \in GL_n(\mathbb{R})(A)$. By number 7, the number of 1s, -1s, and 0s

stays invariant of choice of basis. So $n_a = n_b$, $m_a = m_b$, and $k_a = k_b$. So all matrices in the same orbit have the same number of 1s, 0s, and -1s.

Say A, B have the same number of 1s, 0s, and -1s. $\exists P, Q \in GL_n(\mathbb{R}) \ni P^T A P = \begin{bmatrix} I_n & 0 & 0 \\ 0 & -I_m & 0 \\ 0 & 0 & 0_k \end{bmatrix} = Q^T B Q \implies B = (PQ^{-1})^T A PQ^{-1} \implies B \in GL_n(\mathbb{R})(A)$. $\square$

Claim 13c. $\forall P \in GL_n(F)$, $\dim \ker AP = \dim \ker PA = \dim \ker A$

Proof. $\forall v \in \ker A$, $PAv = 0 \implies v \in \ker PA \implies \ker A \subset \ker PA$. By TKL, $\dim \ker PA \leq \dim \ker P + \dim \ker A = \dim \ker A$. So $\dim \ker A \leq \dim \ker PA \leq \dim \ker A$.

$(AP)_*(V) = \{APv | v \in V\} = A_*(P_*(V)) = A_*(V)$ (since P is invertible). By rank Nullity, $\dim \ker AP = \dim V - \dim (AP)_*(V) = \dim V - \dim A_*(V) = \dim \ker A$. $\square$

Problem 14. Are any two invertible skew symmetric matrices conjugate? Prove they are, or give a simple example to show they need not be.

Proof. Two matrices are conjugate if they have the same eigenvectors.

Take $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$ in $GL_2(\mathbb{C})$.

$P_A(\lambda) = (1 - \lambda)(2 - \lambda) + 1 = \lambda^2 - 2\lambda + 2$ $P_B(\lambda) = (2 - \lambda)(2 - \lambda) + 1 = \lambda^2 - 4\lambda + 5$.

Eigenvalues of A are $1 \pm i$. Eigenvalues of B are $4 \pm 2i$. A and B are not conjugate. $\square$